Antoine Weber
Machin learning

Assignment 3

Part B — Explain your neural network

Q1. Architecture and parameters.

For the baseline model, we have 784 inputs, 32 hidden neurons and 10 output neurons.

The first layer has (784 × 32) + 32 = 25,120 parameters, and the output layer has (32 × 10) + 10 = 330 parameters. So, the model has 25,450 trainable parameters in total.

Parameters are the weights and biases that the model learns during training. Hyperparameters are settings that we choose before training, like the number of neurons, the learning rate or the batch size.

Q2. Activation and loss.

We use ReLU in the hidden layer because it helps the neural network learn more complex patterns in the images.

Softmax is used in the output layer because we have 10 different clothing categories. It gives a probability for each category, and all probabilities add up to 1.

We use sparse categorical cross-entropy because our problem is multiclass classification and the labels are integers from 0 to 9.

Q3. Training and Keras functions.

The model.compile() function is used to choose the optimizer, loss function and metrics. Then, model.fit() trains the model using the training data.

model.evaluate() checks the performance of the model, while model.predict() is used to make predictions on images.

During training, backpropagation calculates the gradients to see how the weights affect the loss. The optimizer then uses these gradients to update the weights and try to reduce the error.

Q4. Interpreting your experiments.

Experiment 1, with 64 neurons, ReLU and Adam, gave the best validation accuracy of 88.23%.

Increasing the number of neurons from 32 to 64 improved the accuracy compared to the baseline (87.33%). Using tanh also improved the result a little (87.77%), but changing Adam to SGD gave a lower accuracy (84.47%).

So, changing the settings does not always improve the model. In our experiments, Adam worked better than SGD with the same number of epochs.

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Part C — Understand classification errors

The confusion matrix shows which clothing categories the model predicts correctly and where it makes mistakes. For example, the model confused 136 shirts with T-shirts/tops, probably because they have similar shapes. The overall accuracy of 86.77% does not show these specific errors, while the confusion matrix helps us understand which categories are more difficult for the model.